9

Aperiodicity: Is It a Good Idea?

9.1 INTRODUCTION

When working with periodic structures, someone will sooner or later ask the question, “Can new, improved, and innovative designs be obtained by variation of the periodicity?”

For example, what would happen if we in an array of simple straight dipoles made every other dipole a little longer and the rest a little shorter? Since the longer dipoles would resonate at a lower frequency and the shorter at a higher, many designers would expect to obtain a staggered tuning effect, possibly leading to a somewhat broader resonance curve and perhaps with a little luck (and “tweaking”!) a flat top. A highly desirable feature indeed.

However, before we delve into a detailed analysis of this very interesting subject, let us briefly present a typical example as shown in Fig. 9.1. It was obtained from the PMM program for normal and 60° angle of incidence.

We first show the resonant curve for an unperturbed array with total element length $2l = 1.5\,\text{cm}$, $d_x = 0.6\,\text{cm}$, and $D_z = 1.7\,\text{cm}$, as seen in the insert of Fig. 9.1. In the same figure we also show the resonant curve for the same array as above but where the length of every other dipole has been increased to 1.65 cm and the rest have been shortened to 1.35 cm as also shown in the insert in Fig. 9.1.

We observe a double resonance as expected but we also observe a null between them. An inexperienced designer might think that the null could be at least “toned down” if the two resonances were moved closer to each other. Typically

Finite Antenna Arrays and FSS, by Ben A. Munk
ISBN 0-471-27305-8

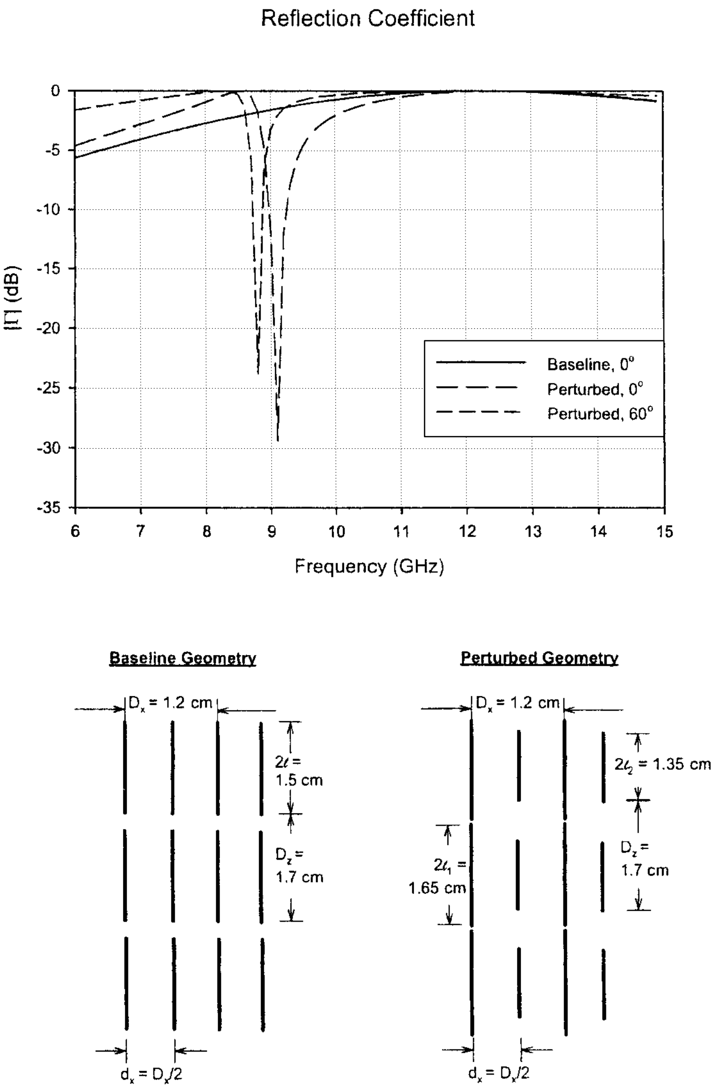


Fig. 9.1 The resonant curve for an array with identical element length (left insert) and also where every other element is longer and the others are shorter (right insert). This may not be particularly useful, but it is interesting.

he would most likely embark on a computational approach backed up by an optimization routine or perhaps just a parametric study. After some praying to the great computer god, he might even arrive at the right conclusion: A flat top without loss just seems impossible for such a configuration. Or maybe we just did not run the program long enough?

However, in the author's opinion a better approach is to analyze the general problem in a classical analytic way. It will be presented in detail below.

But if variation of periodicity in general is somewhat disappointing from an application point of view (not from a theoretical), why spend time on the subject at all? Primarily for two reasons:

1. Even experienced designers sometimes come up with "innovative" designs where the optimizer somehow does not deliver an acceptable design, for the simple reason that none exist.
2. From time to time the periodicity can be disturbed unintentionally. Typically, that might happen if the modes in the PMM calculation are applied improperly. The designer should therefore be familiar with these symptoms if encountered.

However, if the structure is finite, interesting results can be obtained by variation of the periodicity. This case will be investigated in Section 9.6.

We shall next analyze periodic structures in detail when we vary either the element loads, the interelement spacings, or both at the same time.

9.2 GENERAL ANALYSIS OF PERIODIC STRUCTURES WITH PERTURBATION OF ELEMENT LOADS AND/OR INTERELEMENT SPACINGS

The structure to be investigated is shown in Fig. 9.2. It is comprised of two subarrays, each with interelement spacings D_x and D_z. They are interlaced into each other such that array 1 has its reference element located at the origin while the reference element for array 2 is located at $(d_x, 0, 0)$. We denote the element loads by jX_{L1} and jX_{L2} for arrays 1 and 2, respectively. However, we assume that the elements in both arrays have the same length $2l$. That simplifies our analysis by the fact that the self-impedances $Z^{1,1}$ and $Z^{2,2}$ for the two arrays are identical. Thus, the perturbation of the elements is obtained purely by variation of the load resistances jX_{L1} and jX_{L2} and/or d_x. This corresponds approximately to variation of the element length. Furthermore, we denote the mutual impedances between the two arrays by $Z^{1,2}$ and $Z^{2,1}$.

The structure is being exposed to an incident plane wave propagating in the direction $\hat{s}$. It induces the voltages $V^{(1)}$ and $V^{(2)}$ as well as the currents $I^{(1)}$ and $I^{(2)}$ in the two reference elements, respectively.

From the generalized Ohm's Law we thus obtain

$$V^{(1)} = (Z^{1,1} + jX_{L1})I^{(1)} + Z^{1,2}I^{(2)}, \tag{9.1}$$

$$V^{(2)} = Z^{2,1}I^{(1)} + (Z^{2,2} + jX_{L2})I^{(2)}. \tag{9.2}$$

From (9.1) and (9.2) we obtain from Cramer's rule

$$I^{(1)} = \frac{1}{D}[(Z^{2,2} + jX_{L2})V^{(1)} - Z^{1,2}V^{(2)}], \tag{9.3}$$

$$I^{(2)} = \frac{1}{D}[-Z^{2,1}V^{(1)} + (Z^{1,1} + jX_{L1})V^{(2)}], \tag{9.4}$$

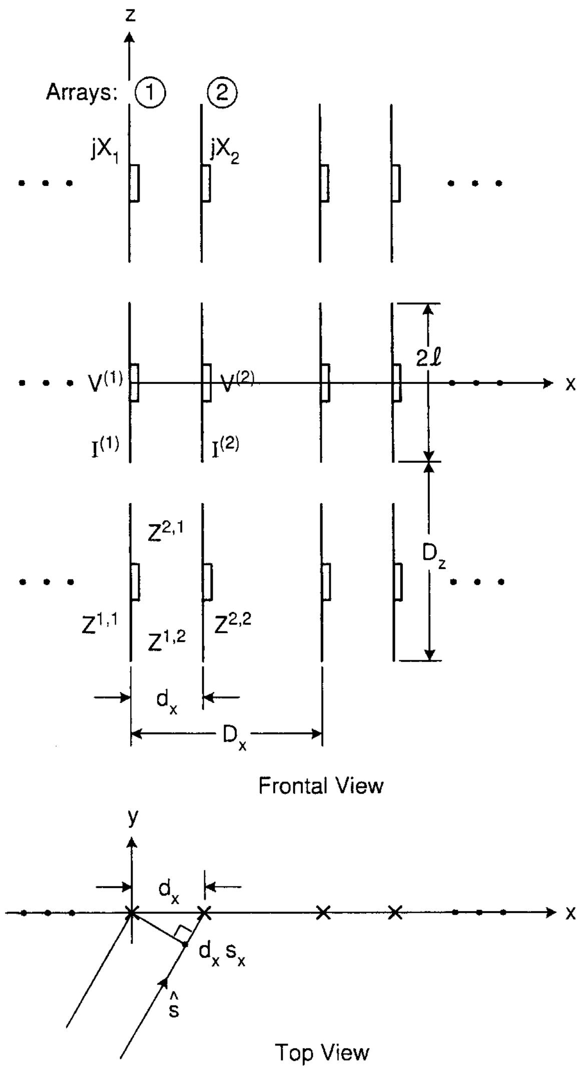


Fig. 9.2 Two arrays 1 and 2 with interelement spacing D_x and D_z interlaced into each other with array spacing d_x. Array 1 is loaded with jX_{L1} and array 2 with jX_{L2}. The element lengths of both arrays are identically equal to 2l.

where the determinant D is

$$D = \begin{vmatrix} Z^{1,1} + jX_{L1} & Z^{1,2} \\ Z^{2,1} & Z^{2,2} + jX_{L2} \end{vmatrix}. \tag{9.5}$$

Furthermore, we assume that the plane of incidence is the xy plane while the incident field $\overline{E}^i$ is oriented along the z axis. Then the voltage induced in the reference element for array 1 is then given by [111]

$$V^{(1)} = E^i P^t(0,0), \tag{9.6}$$

where $P^t(0,0)$ is the pattern factor under transmitting condition for $(k,n) = (0,0)$.

Furthermore, the voltage $V^{(2)}$ induced in the reference element for array 2 has the same amplitude as $V^{(1)}$ (the elements are identical), but the phase is adjusted for its displacement d_x, that is,

$$V^{(2)} = V^{(1)} e^{-j\beta d_x s_x}, \tag{9.7}$$

The element currents $I^{(1)}$ and $I^{(2)}$ are now completely determined by equations (9.3) through (9.7).

Furthermore, we recall that the propagating field $\overline{E}^{(j)}(\overline{R})$ for a single array is given by [63]

$$E^{(j)}(\overline{R}) = I^{(j)}(\overline{R}^{(j)}) \frac{Z}{2D_x D_z} \frac{e^{-j\beta(\overline{R}-\overline{R}^{(j)})\cdot\hat{r}_\pm}}{r_y} \overline{e}_\pm(0,0) P^{(j)}(0,0), \tag{9.8}$$

for no grating lobes, where $\overline{R}^{(j)}$ denotes the reference point for array "j," $\overline{e}_\pm = (\hat{z} \times \hat{r}_\pm) \times \hat{r}_\pm$ (see [63]), and $P^{(j)}$ is the pattern factor under scattering conditions. The total bistatic scattered field from the two subarrays is now obtained by addition of the fields from the two arrays given by (9.8), just like an ordinary antenna problem.

Substituting the values for $I^{(1)}$ and $I^{(2)}$ as calculated above and noting that

$$R_A = \frac{Z}{2D_x D_z} \frac{1}{s_y} P(0,0) P^t(0,0), \tag{9.9}$$

we readily obtain the following equation for the total reflection coefficient for the two arrays:

$$\Gamma = -R_A \frac{(Z^{1,1} + jX_{L1}) - Z^{1,2} e^{-j\beta d_x s_x} - Z^{2,1} e^{+j\beta d_x s_x} + (Z^{2,2} + jX_{L2})}{(Z^{1,1} + jX_{L1})(Z^{2,2} + jX_{L2}) - Z^{1,2} Z^{2,1}}. \tag{9.10}$$

9.2.1 No Grating Lobes from Subarrays

When no grating lobes from the subarrays exist, we have

$$Z^{1,1} = Z^{2,2} = R_A + jX_A. \tag{9.11}$$

Furthermore, based on the general plane wave expansion, Hill has obtained a general expression for the mutual impedances [112]:

$$Z^{1,2} = (R_A - R_M + jX_M) e^{+j\beta d_x s_x} \tag{9.12}$$

for no grating lobes and

$$Z^{2,1} = (R_A + R_M + jX_M)e^{-j\beta d_x s_x} \tag{9.13}$$

for no grating lobes. Note that $R_M = 0$ for $d_x = D_x/2$ and also for normal incidence. However, we always have $Z^{1,2} \neq Z^{2,1}$ except if the two subarrays are displaced right behind each other—that is, along the y axis only.

After straightforward but somewhat tedious calculations, substituting (9.11), (9.12), and (9.13) into (9.10) yields

$$\Gamma = -\frac{1}{1 + j\dfrac{X_A^2 - R_M^2 - X_M^2 + X_A(X_{L1} + X_{L2}) - X_{L1}X_{L2}}{R_A(2X_A - 2X_M + X_{L1} + X_{L2})}}. \tag{9.14}$$

Without any loss of generality except a simple frequency shift, we now assume asymmetric loading (i.e., $X_{L1} = -X_{L2} = X_L$), which reduces (9.14) to

$$\Gamma = -\frac{1}{1 + j\dfrac{X_A^2 - R_M^2 - X_M^2 - X_L^2}{2R_A(X_A - X_M)}}. \tag{9.15}$$

Inspection of (9.15) readily shows that we obtain $\Gamma = -1$ (resonance) for

$$X_A = \pm\sqrt{R_M^2 + X_M^2 + X_L^2} \tag{9.16}$$

and a null ($\Gamma = 0$) for

$$X_A = X_M. \tag{9.17}$$

In Fig. 9.3 we show a typical general resonance curve as a function of X_A for a structure comprised of two subarrays with reactive loads $jX_{L1} = -jX_{L2} = jX_L$ and with interelement spacing D_x and D_z. The reference elements of the second array are shifted d_x along the x axis only.

Also shown is the baseline curve for $d_x = D_x/2$ and $X_L = 0$.

Note that since $R_M = 0$ in that case, we obtain at resonance from (9.16) $X_A = \pm X_M$. However, only the negative value is correct since the positive value is canceled by the null at $X_A = +X_M$ [see (9.17)].

Based on the plane wave expansion, Hill derived an expression for X_M [112]. He found that for a straight dipole, X_M would in general be negative (it appears that positive X_M could only occur if the current distributions were odd; for tripoles, see Section 9.3).

Thus, in Fig. 9.3 the baseline resonance at $-X_M$ will be located to the right of $X_A = 0$. Furthermore, by inspection of (9.16) we observe that the highest resonance is always higher than the baseline resonance at $X_A = -X_M$, depending on X_L and R_M.

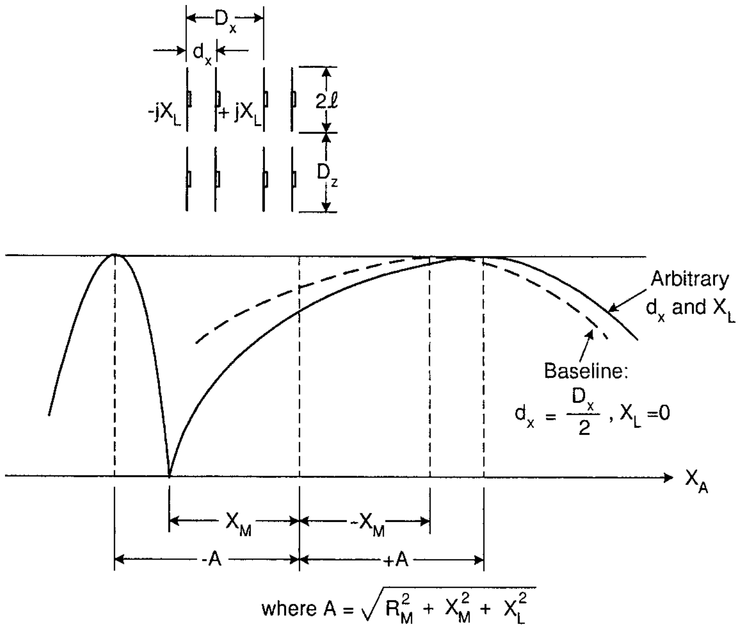


Fig. 9.3 The resonance curve as a function of X_A for two arrays with interelement spacings D_x and D_z interlaced into each other with array separation d_x. The element lengths are the same for both arrays, but one array is loaded with jX_L and the other with $-jX_L$. Note that regardless of spacing $d_x \neq D_x/2$ and $jX_L \neq 0$, we always obtain a null at $X_A = X_M$ and a main resonance slightly higher than the unperturbed configuration.

Furthermore, the lowest resonance is always lower than the null at $X_A = X_M$; that is, the null will always be located between the two resonances, in agreement with Foster's reactance theorem [101].

It is also interesting to note that the null is independent of d_x as well as X_L while the two resonances depend strongly on both of these parameters.

Finally note that the highest resonance for the perturbed case is as much above the baseline resonance at $X_A = -X_M$ as the lowest resonance frequency is below the null at $X_A = X_M$.

9.2.2 Grating Lobe from Subarrays

So far we have assumed that the interelement spacings D_x and D_z were so small that no grating lobes would emerge from the two subarrays. However, if this is not the case, the expressions above must be modified. More specifically, (9.11) becomes

$$Z^{1,1} = Z^{2,2} = R_A + R_G + jX_A \tag{9.18}$$

when grating lobes are present. R_G denotes a resistance representing the extra energy disappearing in the direction of the grating lobe(s).

Similarly, if the elements are regularly spaced (i.e., $d_x = D_x/2$), (9.12) and (9.13) should according to Hill [112] be modified to

$$Z^{1,2} = (R_A - R_G + jX_{MG})e^{+j\beta d_x s_x} \tag{9.19}$$

for grating lobes and modified to

$$Z^{2,1} = (R_A - R_G + jX_{MG})e^{-j\beta d_x s_x} \tag{9.20}$$

for grating lobes. If unevenly spaced, (9.19) and (9.20) become more complicated.

Substituting (9.18), (9.19), and (9.20) into (9.10) yields the reflection coefficient Γ where grating lobe(s) can emerge from the two subarrays as evaluated by Hill [112].

However, in this chapter it will be sufficient to state the overall effect of grating lobes:

1. The upper resonance—that is, the one slightly higher than the resonance of the unperturbed structure—will exhibit a slight loss due to the extra energy that must be diverted to the grating lobe directions. The loss typically ranges from a few tenths to a few decibels.
2. The lower resonance as well as the null in the resonance curve will be more or less washed out, depending on the onset frequency of the grating lobes.

9.2.3 Concluding Remarks for Section 9.2

Variation of the inter element spacing and/or the reactive tuning loads of a periodic structure will result in a new resonance slightly above the unperturbed resonance and without loss, provided that no grating lobes exist. If grating lobes from the two subarrays do exist, we will experience loss at resonance.

There will also be a lower and narrower first resonance without loss, provided that no grating lobes exist. A perfect null is located between the two resonant frequencies. It too will partly disappear when grating lobes occur.

Thus, a resonance curve with a flat top and without loss appears not to be obtainable from a single surface. However, it is possible if the two arrays are cascaded behind each other (see reference 113).

9.3 PERTURBATION OF ARRAYS OF TRIPOLES

We have so far considered perturbations of arrays of just straight parallel dipoles. Although they are indicative of what may happen when we apply more complicated elements instead, we shall now specifically investigate arrays of tripoles. (Taken from Hill's dissertation [112].)

From the PMM program we obtain the resonance curve for an array of tripoles arranged in a regular equilateral triangular grid as shown in Fig. 9.4

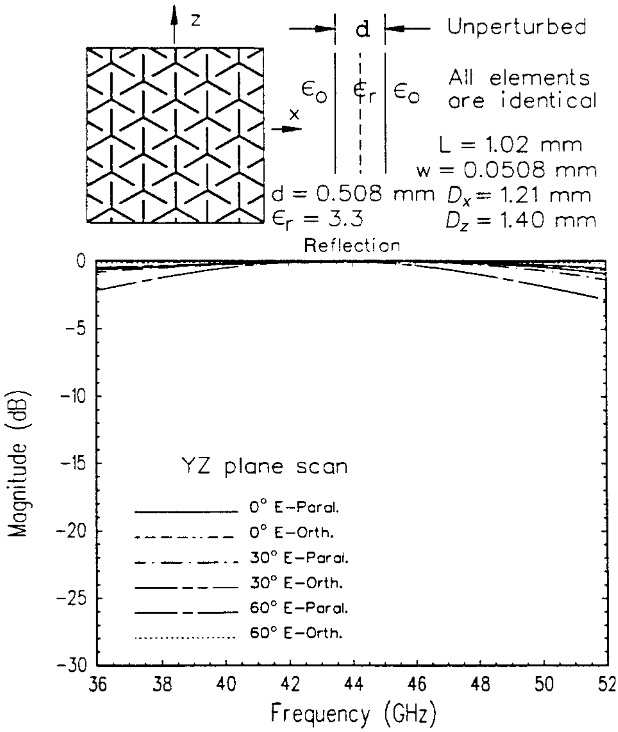


Fig. 9.4 Calculated reflection response of a tripole array with no perturbations.

(our baseline). We next show in Fig. 9.5 the same array except that each fourth element has been reduced 20% in size. Similarly we show in Fig. 9.6 the case where the same elements have been increased by 20% instead. Inspection of these curves clearly shows that the shorter elements push the resonance frequency of the array upwards and vice versa for the longer case. While that in itself is both expected as well as acceptable, we also observe a number of deep dips in the resonance curve that would make such an arrangement less than desirable. Finally in Fig. 9.7 we show the extreme case where each 4*th* element is completely missing. Although that configuration could be used for some less demanding application, it is obviously not an approach that should be pursued seriously, not even if we observe no loss at resonance in spite of the fact that we have fewer elements. In other words, to obtain a quality periodic surface, we must observe periodicity in the strictest sense.

9.4 MAKING USE OF OUR OBSERVATIONS

The investigation above can be highly instructive. We shall illustrate this statement by several examples.

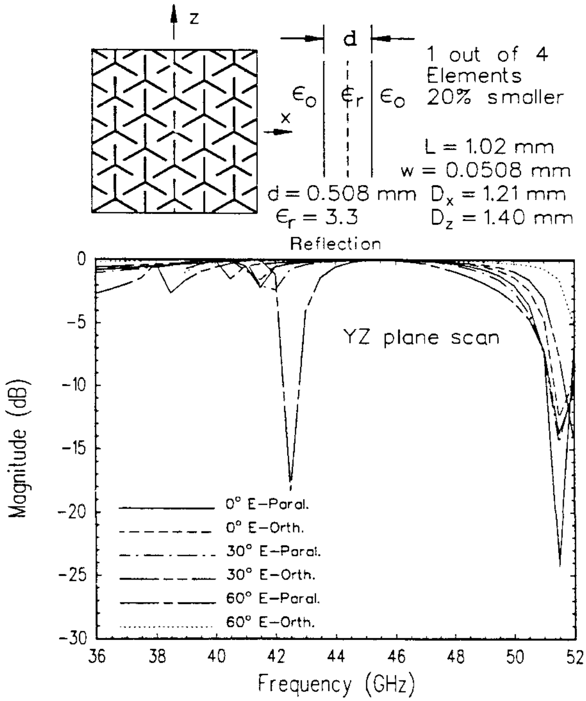


Fig. 9.5 Calculated reflection response of a tripole array with every fourth element 20% smaller.

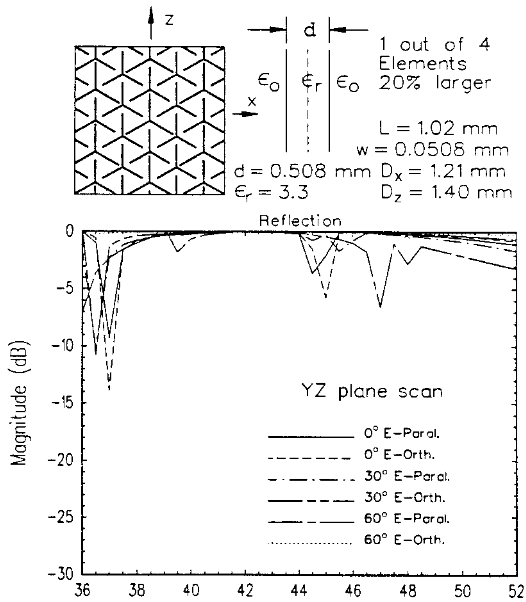


Fig. 9.6 Calculated reflection response of a tripole array with every fourth element 20% larger.

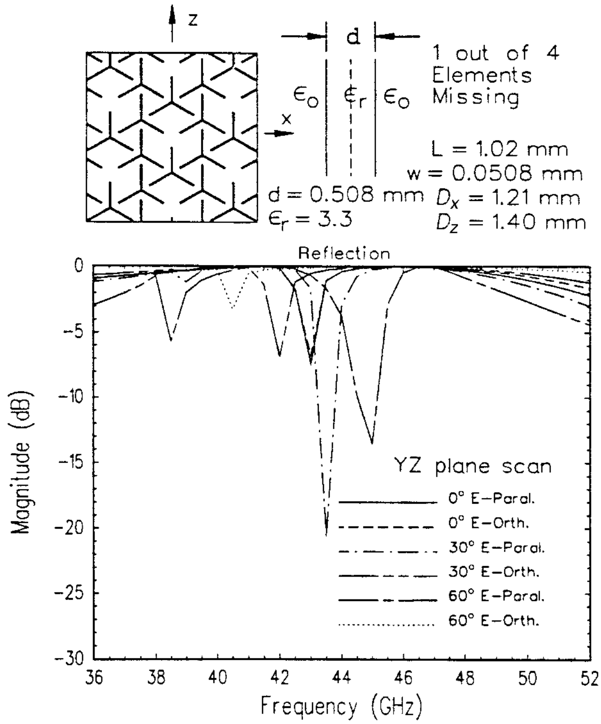


Fig. 9.7 Calculated reflection response of a tripole array with every fourth element missing.

9.4.1 Multiband Designs

Every so often, dual-band designs are called for—as, for example, two resonances separated by a null or valley. The approach to this problem depends to a high degree on the ratio between the two bands. If it is low, one possibility is to use concentric loops as illustrated in Fig. 9.8. Recalling that the circumference of a resonating loop element is about one wavelength, we readily conclude that the largest loop should have a diameter $\sim\lambda_L/3$, which also is the smallest interelement spacing in a rectangular grid. At the high frequency the interelement spacing should be no more than $\lambda_H/2$ to avoid grating lobes for all scan angles, that is, $f_H < 1.5 f_L$. This constitutes about the largest frequency ratio that can be used for concentric loops. However, if we instead of loops use hexagons embedded in dielectric, the frequency ratio could be driven up to as high as $\sim$1.8:1.

If higher frequency ratios are called for, we must resort to other approaches.

One possibility shown in Fig. 9.9 is comprised of large Jerusalem crosses interlaced with smaller ones (not recommended, read on). At the lowest frequency only the large elements will be strongly excited, yielding as good a performance as one can expect of the Jerusalem cross (the small elements are only weakly excited).

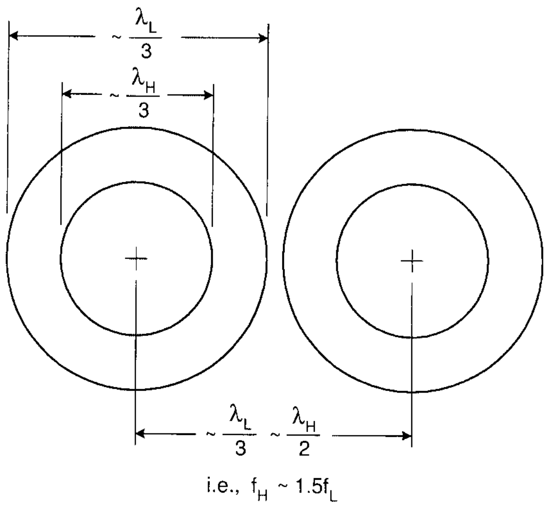
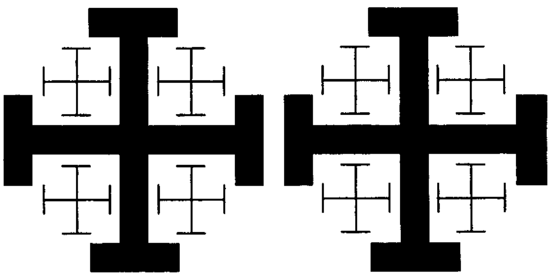


Fig. 9.8 An array of concentric loops will have two resonances. However, the ratio between the two should not exceed 1.5:1 unless "distorted" into hexagons and embedded in dielectric. See text.



Not recommended !

Fig. 9.9 An array of large Jerusalem crosses interlaced with small ones. In the low-frequency band, only the large elements will be excited, resulting in satisfactory performance. However, in the high-frequency band the small element will be in an irregular environment, resulting in mediocre performance in general.

At the higher frequency band there is always the possibility that a higher resonance will be excited on the larger elements—in particular, at an oblique angle of incidence. However, even if we luck out in that respect, we do observe that the "periodicity" of the small elements has been violated on several accounts. For example, one end of the small elements is close to the large elements, while the other end is not. Thus, even if the small elements were arranged with the "proper" element spacings, the uneven end loading would produce undesirable nulls, similar to what is seen for example in Fig. 9.3. Thus, the design shown in Fig. 9.9 is not recommended.

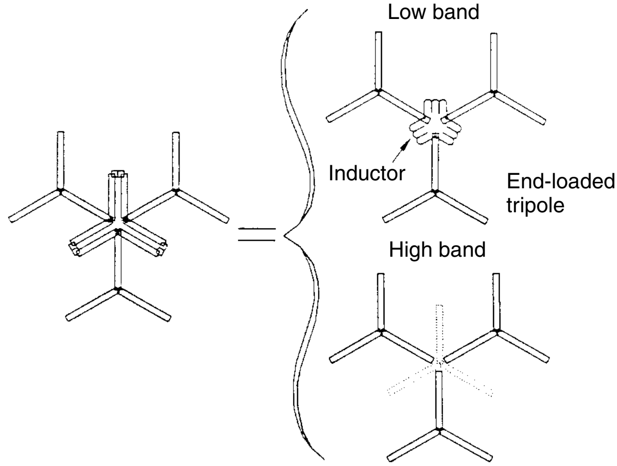


Fig. 9.10 A sketch of the original "snowflake" element showing the equivalent structures for the low and high bands. Each leg in the three-legged trap is about $\lambda_H/4$ long, leading to separation of the three tripoles.

9.4.2 The Original "Snowflake" Element

We saw above how seemingly sound arrangements of elements were deemed undesirable because the periodicity was violated. An attempt to rectify this situation is to use an alternative element—as, for example, the snowflake element shown in Fig. 9.10. It is basically made of three tripoles joined together by three traps in parallel and each having a length of $\sim\lambda_H/4$ at the high frequency. Thus the three tripoles are basically disconnected at the high frequency band while they are joined together by inductive loading at the low frequency band such that they basically form an end-loaded and center-loaded tripole. Furthermore, the three-armed trap acts like the fourth tripole at the high band that should be there in order not to disturb the periodicity. A typical performance is shown in Fig. 9.11. We observe satisfactory performance at normal angle of incidence in both bands and also for oblique incidence in the low band as well. However, the E-plane polarization at the high band is not satisfactory.

One of the fundamental problems with this type of element is that the optimum length when acting as traps may not coincide with the correct resonant frequency of the mid-element. And as we saw in Figs. 9.5 and 9.6, this leads to undesirable resonances. Some improvement in that respect can be obtained by modification of the original snowflake element as will be described next.

9.4.3 The Modified "Snowflake" Element

The modified snowflake element is shown in Fig. 9.12. We use three traps as earlier, but this time connected to the three tripoles via series capacitors as shown in the insert. This gives us an extra degree of freedom. We typically start the

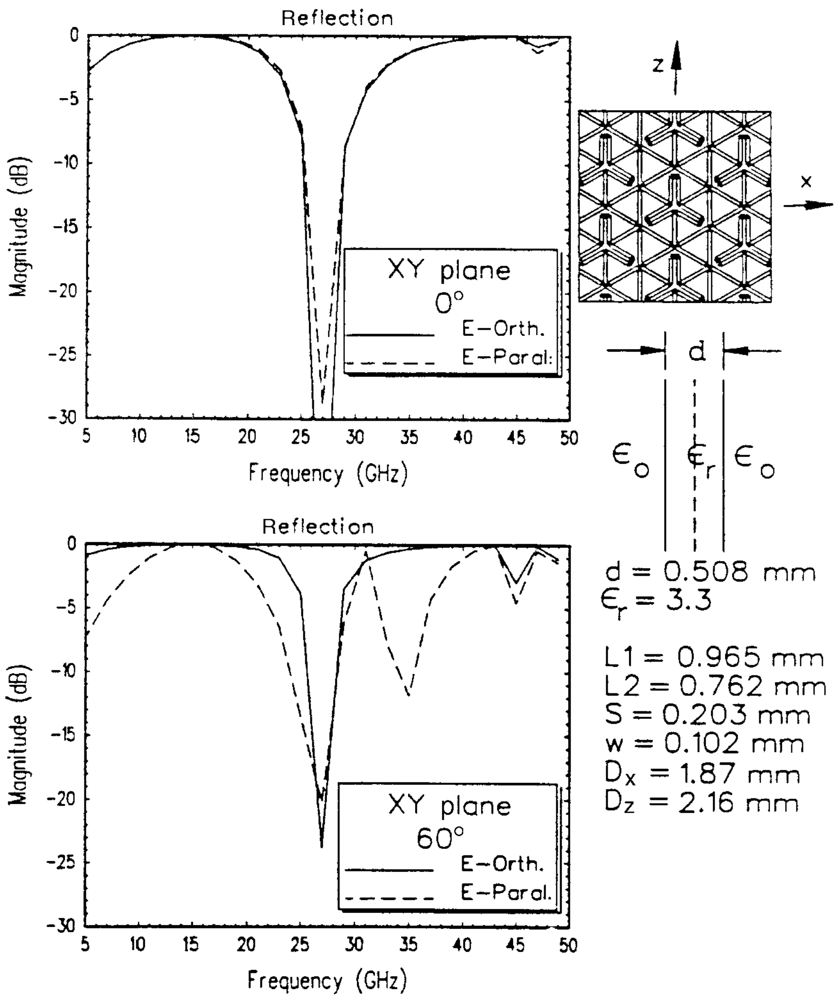


Fig. 9.11 Calculated response for an array of original "snowflake" elements at normal and oblique incidence.

design procedure by choosing the tripoles to resonate at the high-frequency band. Next, we design the traps to also isolate at that frequency. And finally we adjust the capacitors such that we obtain resonance in the low band.

Typical calculated resonant curves for both the low and high bands are shown in Fig. 9.13. The low band is seen to be entirely satisfactory while the high band barely "squeaks" by in a very narrow frequency range. The problem is simply that the three traps do not perform well as a fourth element at the high band. Consequently, we will experience the type of resonances illustrated in Figs. 9.5, 9.6, and 9.7. The element has been deemed satisfactory when used in radomes for communication purposes. We would not recommend it for precision purposes.

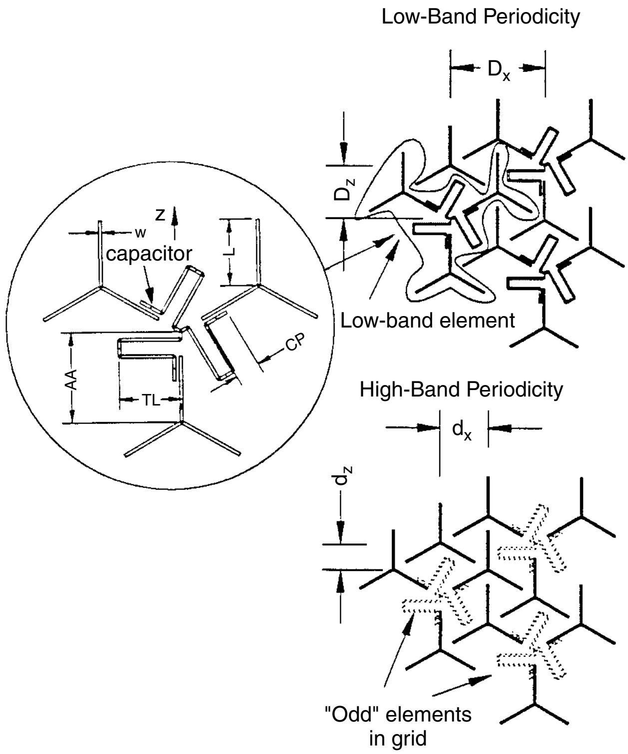


Fig. 9.12 Demonstration of the periodicity of the modified "snowflake" element. Consists of three tripoles joined together with traps with series capacitors.

9.4.4 Elements with Mode Suppressors

At this point the reader is entitled to a design that really works! Well, here it is.

It was demonstrated above that periodicity must be preserved and the interelement spacing be small to obtain stability of the resonant frequency with angle of incidence. Fundamentally, that requires elements that are not only small but also of such a shape that close packing is possible. Tripoles are particularly strong candidates in that respect. However, we must also make sure that the low-band elements do not have a second resonance somewhere in the high band. And this is precisely where the tripole has a problem if the frequency ratio is around 2:1. In fact, by inspection of Fig. 2.10 in reference 114 we observe a typical strong second resonance for oblique angle of incidence in the *E* plane. In other words, it may be desirable to move the fundamental and the second harmonic somewhat apart from each other.

One approach is to use so-called suppressors as shown in Fig. 9.14 in the case of a simple straight dipole. We show the current and voltage distribution for the fundamental (top) as well as the second harmonic (bottom). If we place capacitors either in the form of plates or just wires at the ends of the dipoles

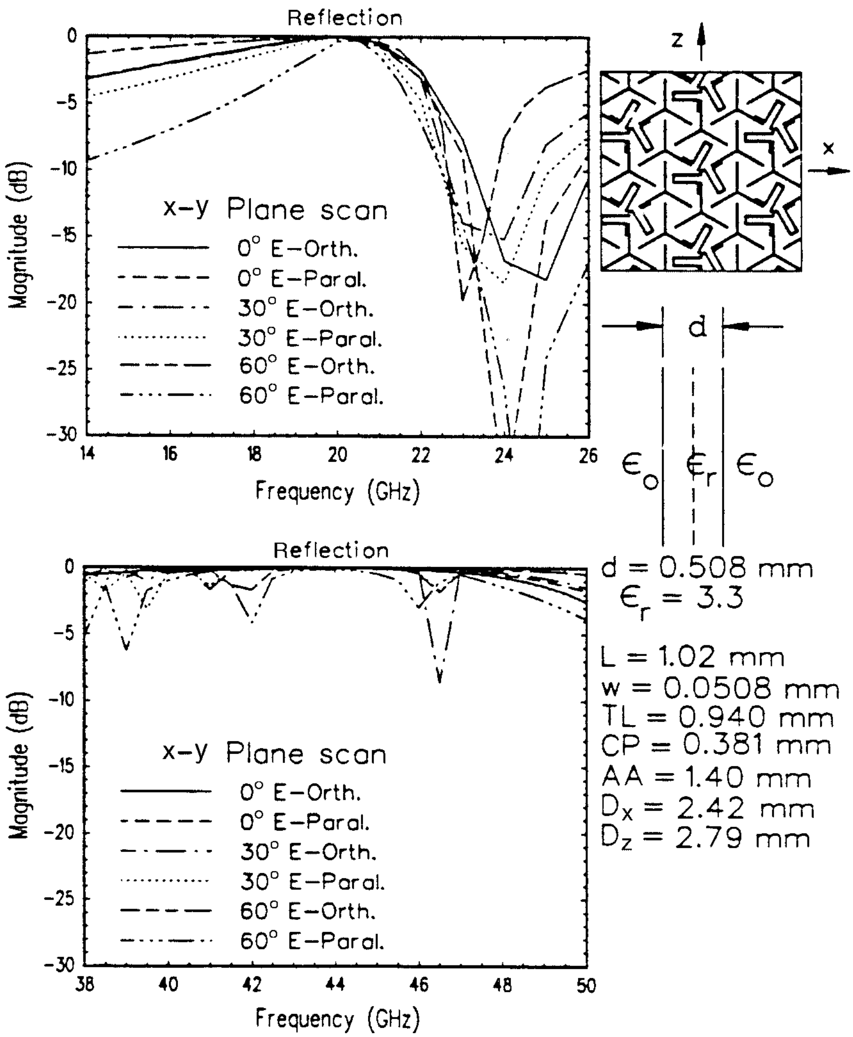


Fig. 9.13 Calculated reflection curves for an array of modified "snowflake" elements in the low (top) as well as the high (bottom) bands.

when the voltage is maximum, it is clear that the fundamental as well as harmonic resonances will be moved strongly downward in frequency.

However, if we place the capacitors at points on the dipole where the voltage for the second harmonic is zero, no change of resonant frequency will take place for this mode. But we will see a considerable shift downward for the fundamental.

The net effect is that we have increased the ratio between the harmonic and the fundamental to be significantly larger than 2:1 as desired.

An example of a dual-band structure made of large tripoles interlaced with small ones is shown in Figs. 9.15 and 9.16. The large tripoles are provided with

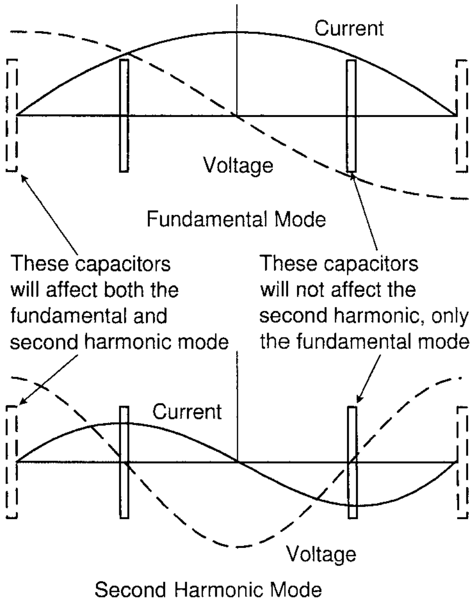


Fig. 9.14 Current and voltage distributions along a straight dipole for the fundamental and second harmonic modes. Capacitive plates placed at the ends will detune both modes downward in frequency. However, if placed where the voltage of the second harmonic is zero, no change in resonant frequency will take place. However, the fundamental mode will still be tuned somewhat downward.

capacitive suppressors as described above, and the periodicity of both kinds of elements has been maintained rigorously. Scans in both the *YZ* and the *XY* planes are shown. The results speak for themselves (well done Roger!).

9.4.5 Alternative Design

We saw in the previous section how the fundamental and the second harmonic of a tripole element could be moved apart from each other by use of suppressors. We could in that way avoid excitation of the second harmonic of the low-frequency elements. However, by inspection of the resonance curve for the loaded three-legged element shown in Fig. 2.23 in reference 114 we observe a distinct null at a frequency twice the fundamental. Thus, the tripole loop element is already suitable for a dual-band surface with the ratio 2:1 between the two bands.

Although this idea has not been tried (it emerged after Roger H. graduated), the author would be very much interested in hearing why it would not work.

9.5 ANOMALIES DUE TO INSUFFICIENT NUMBER OF MODES (THE UNNAMED ANOMALY)

In this section we shall present an example of an anomaly caused by insufficient number of modes on the elements.

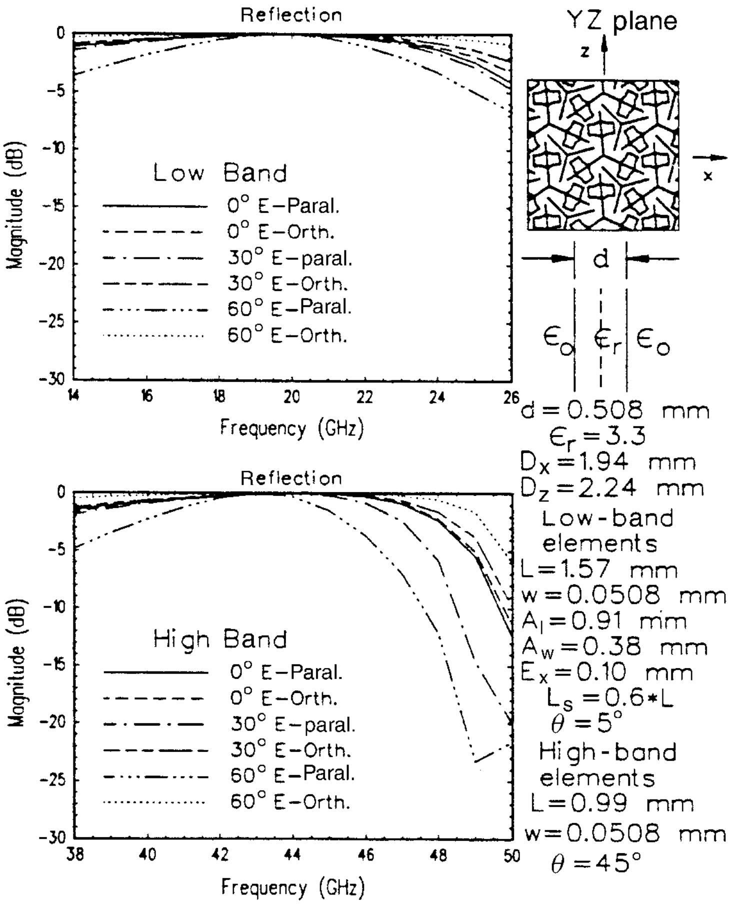


Fig. 9.15 Calculated low and high band YZ plane response for a mid-arm loaded multiple tripole element array.

Consider the array of four-legged loaded elements shown in Fig. 9.17. As indicated in the insert to the right, we have used four modes to obtain the resonant curve shown for oblique angle of incidence ($\eta = 45°$). However, if the incident E field is vertically polarized, it appears that the current across the top and bottom of the elements should be zero. Thus, one could argue that only the two modes shown in the insert to the left are sufficient. However, although the resonance curve in that case basically "agrees" with the four-mode case, we do observe a particular anomaly around $f = 7.2$ GHz. What has happened?

Well, in the two-mode case the two modes simply act as the reference elements in two interlaced arrays with interarray spacing $d_x \neq D_x/2$. Thus, we will obtain an effect analogous to the phenomena shown in Fig. 9.3. Not before we add the connector modes at the top and bottom will the two arrays be joined into *one* array with the proper phases on the left and right halves of the elements.

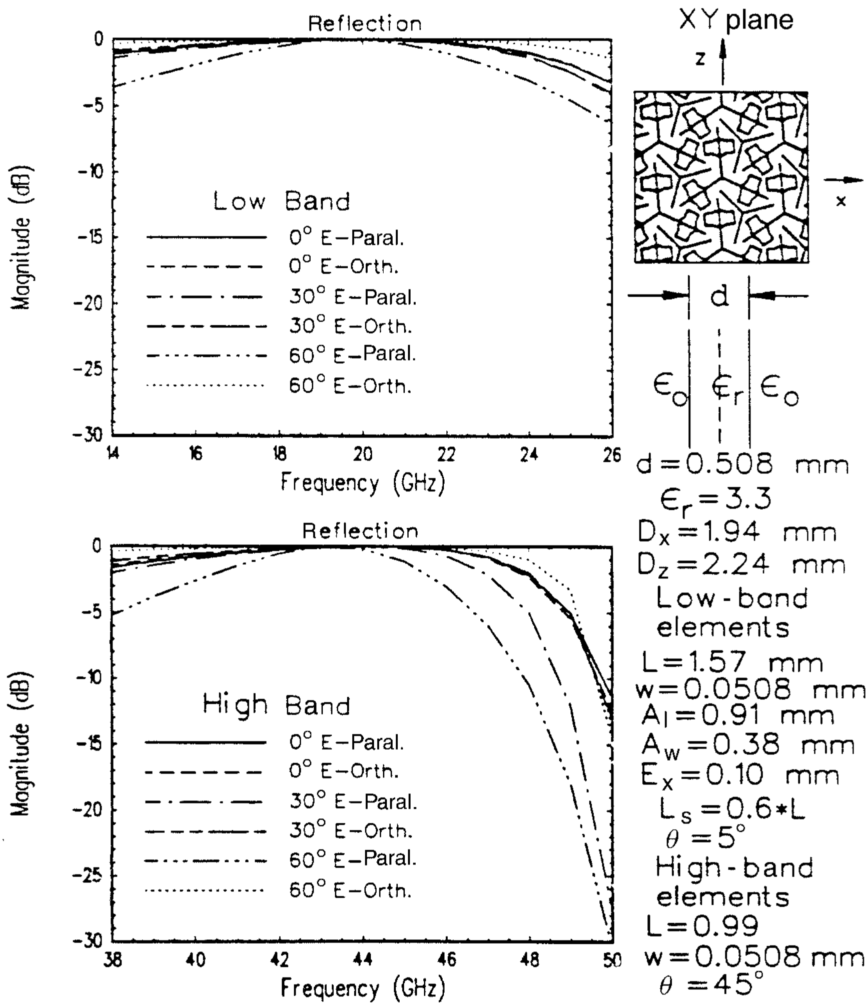


Fig. 9.16 Calculated low- and high-band XY plane response for a mid-arm loaded multiple tripole element array.

Thus, even if computer time can be saved, don't be too stingy with the number of modes. On the other hand, putting on too many can lead to problems too, as discussed in reference 115.

Note that no anomaly of this type is observed for normal angle of incidence.

This anomaly was first observed more than a dozen years ago. A highly respected researcher and friend at first refused to believe the explanation given above. To tease him, the rest of us named this anomaly after him. He eventually asked that his very good name be "crossed out." Actually, he should not be embarrassed: This is the one and only time I can ever remember where he "missed it."

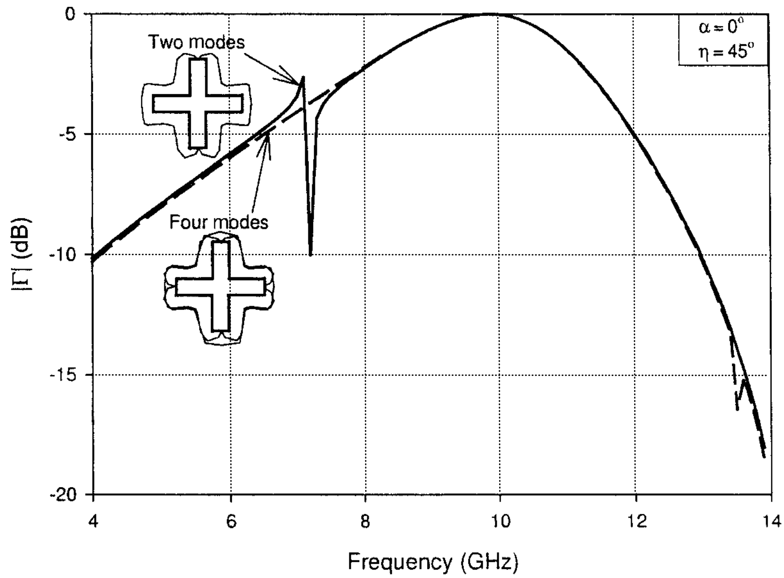


Fig. 9.17 The resonant curves for a single array of four-legged loaded elements at 45° angle of incidence in the H plane. One curve shows the correct response when using four modes (minimum). The other curve (full line) shows the response when using only two modes corresponding to two interlaced arrays with d_x different from $D_x/2$ as shown in Figure 9.3.

9.6 TAPERED PERIODIC SURFACES

9.6.1 Introduction

A tapered periodic surface (TPS) is an array of wire or slot type elements with element sizes and interelement spacings that vary as a function of position over the array[1]. Many different applications of TPS come to mind. Most of these applications involve a smooth electrical transition to terminate the finite edge of an FSS or phased array. A TPS could also be used to transition from one FSS design to another in a radome with multiband window areas. Another application is as a building block for broad band and low side lobe antenna designs. Tapered Periodic Surfaces exhibit two fundamental properties:

1. **Diffraction Control:** A TPS provides a tapered impedance surface similar to a tapered resistive surface or tapered R-card. The difference is that a TPS is a tapered reactive surface and could be thought of as a tapered "jX" card.
2. **Frequency Compensation:** A TPS effectively behaves as an electrical conductor that changes size as a function of frequency. This property

[1] English, Errol K, and Leeper, William J., "Periodic Surfaces for Selectively Modifying the Properties of Reflected Electromagnetic Waves," United States Patent Number 5,606,335 Patent Date February 25, 1997.

enables the design of radiating elements and/or apertures that become effectively smaller with increasing frequency. Thus a TPS maintains a constant electrical size, resulting in electrical performance that is very constant over a large bandwidth.

9.6.2 Physical Description

The majority of this discussion will be limited to tapered periodic surfaces used to smoothly terminate a finite edge of a periodic surface. Typically, the element sizes become progressively (monotonically) smaller along the length of the taper. The periodic surface elements may be of any type: linear, four-legged-unloaded, four-legged-loaded, three-legged, etc. The elements may be spaced in a relatively sparse grid or an extremely dense grid depending on the application. A TPS is normally produced by printed circuit methods where the elements reside on a dielectric substrate.

Slot and Wire Type TPS Figure 9.18 shows a TPS of simple straight wire segments. This example is referred to as a parallel type because the wire segments are parallel to the direction of the taper. The edge with the long elements is the baseline or low impedance edge while the edge with the short elements is the terminal or high impedance edge. In an analogous manner, an orthogonal type TPS has wire segments oriented orthogonal to the direction of taper. Note that the TPS of Fig. 9.18 is designed only for linear polarization. If a wire TPS for arbitrary polarization is desired, an element type such as three legged or four legged might be used rather than the simple linear elements. However, the greatest bandwidth is obtained by very tight packing of the elements in the grid, and straight wire elements can be packed extremely tight. Therefore, the largest bandwidth (for arbitrary polarization) is achieved by superimposing a parallel type and an orthogonal type TPS, one behind the other.

In the case of a slot TPS, the edge with the short elements is the baseline or low impedance edge while the edge with the long elements is the terminal or

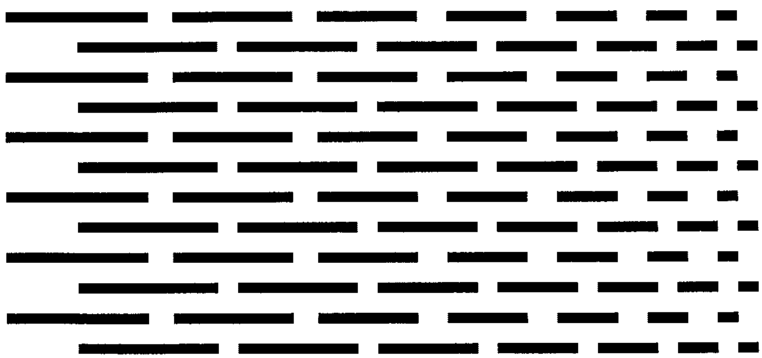


Fig. 9.18 Parallel type Tapered Periodic Surface of linear wire segments.

high impedance edge. As in the case of the wire TPS, the parallel and orthogonal descriptions refer to slot element orientation relative to the direction of taper. An orthogonal slot TPS and a parallel slot TPS may not be superimposed one behind the other. The use of one precludes the other. If a taper for arbitrary polarization is desired, slot elements such as three legged or four legged must be used in a single surface.

General TPS dimensions depend greatly on the specific application and the frequency range of operation. Typically, for applications in the microwave frequency band (2 to 18 GHz), the length of the taper may be anywhere from 2 to 24 inches. Also, the element width and the gap width are typically in the vicinity of 0.002 inches to 0.020 inches for broad band applications.

The length of the longest segments of the TPS is typically chosen to be approximately $\lambda/2$ or less at the center of the operating frequency band. The length of the shortest segments at the terminal edge is chosen to be a vanishingly small fraction of a wavelength at the highest operating frequency.

The thickness of the substrate supporting the TPS is application dependent but is typically between 0.002 inches to 0.020 inches. Good substrate materials have a fairly low dielectric constant and loss tangent. Typical substrate materials include glass/epoxy (such as FR4), glass/PTFE, quartz/polyimide, quartz/cyanate-ester, as well as kapton or mylar film.

9.6.3 Purpose and Operational Description

Diffraction Control One purpose of a tapered periodic surface is to provide a gradual transition from a good conductor (i.e. metal) to free space. Abrupt termination of a conducting edge gives rise to a very strong diffracted field when illuminated by an externally impressed EM wave. A tapered periodic surface provides a gradually tapered surface impedance, eliminating the abrupt termination and thereby significantly reducing diffracted fields. Figure 9.19 illustrates the use of a TPS at a metal edge.

Functionally, a tapered periodic surface performs in a fashion similar to a tapered resistive film (sometimes referred to as an R-card or edge card). Ideally,

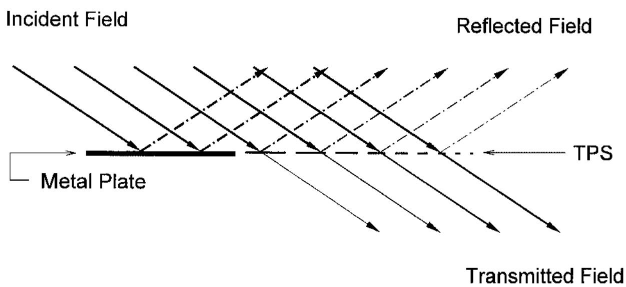


Fig. 9.19 An edge-view of a Tapered Periodic Surface used to gradually transition from metal to free space.

a tapered resistive film transitions impedance from $Z = 0\,\Omega/\text{sq}$ to $Z = \infty\,\Omega/\text{sq}$ assuming real (resistive) values. TPS tapers provide similar impedance transitions assuming imaginary (reactive) values. A wire TPS transitions from $Z = 0\,\Omega/\text{sq}$ to $Z = -j\infty\,\Omega/\text{sq}$ along a path of increasing capacitive reactance. A slot TPS transitions from $Z = 0\,\Omega/\text{sq}$ to $Z = +j\,\infty\Omega//\text{sq}$ along a path of increasing inductive reactance.

A periodic surface of closely spaced wire segments will be highly reflective (almost identical to solid metal) over a wide frequency range centered at the frequency where the segments are approximately a half wavelength long. When the periodic surface is highly reflective, its surface impedance is very close to $0\Omega/\text{sq}$ (the same as a perfect conductor). This is the situation at the baseline edge of a wire TPS. As the wire segments are gradually shortened (along the length of the taper), the wire segments are increasingly below resonance and the surface impedance (purely capacitive reactance) increases, and the electromagnetic wave reflection gradually decreases as the transmission gradually increases. This gradual transition from total reflection to total transmission eliminates the hard reflection and transmission shadow boundaries and therefore results in significantly reduced levels of diffracted fields.

The following example illustrates an example of diffraction control using a wire TPS. A parallel type TPS of straight wire elements was designed for broad band use (2 to 18 GHz). At the baseline edge, the elements are 0.295 inches long, 0.002 inches wide, and spaced in a skewed grid with a side-by-side periodicity of 0.004 inches. The lengths of the elements are linearly tapered to zero over a 12-inch distance. Figure 9.20 shows this TPS attached to the edge of a solid sheet of copper. The solid copper and the TPS are supported by a thin glass/epoxy substrate and backed by foam (0.25 inches thick). This is incorporated into a test article for radar cross section (RCS) testing.

The RCS testing was conducted with the illumination at normal incidence to the metal edge. Also, the polarization is parallel to the plane of incidence (orthogonal to the metal edge, i.e. the hard diffraction case). Figure 9.21 shows the monostatic RCS at 30° above grazing for the TPS attached to the trailing metal edge. The RCS of an abruptly terminated trailing metal edge is also shown in Figure 9.21 for comparison. Note that the TPS reduces the diffracted field in the monostatic direction by approximately 25 dB over the entire 2 to 18 GHz band.

Frequency Compensation Frequency compensation is a very important property of the TPS that is not shared by the tapered R-Card. Since the TPS is a periodic surface, it is frequency sensitive, a property that may be exploited on a variety of broad band antenna applications.

If we think of a TPS as being highly opaque (conductive) at one end and highly transparent (nonconductive) at the other end, then there is some point in between where the transmission is at its halfway point or 3 dB point. The position along the taper of this 3 dB point varies with frequency. Therefore, we may conclude that the TPS effectively behaves like a conductor from its conductive end to the 3 dB point of the taper. The equivalent conductive length of the TPS varies monotonically with frequency. This is true for both wire and slot TPS.

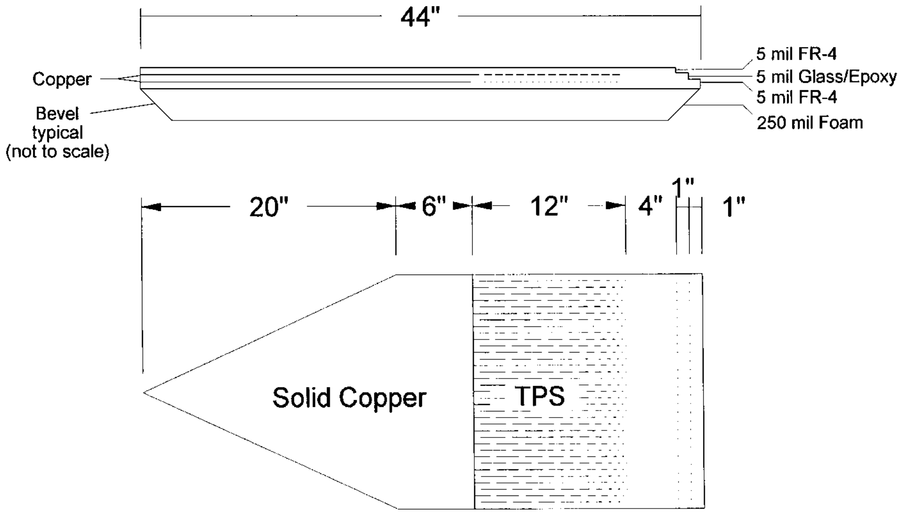


Fig. 9.20 RCS test article used to evaluate the diffraction reduction effectiveness of a TPS.

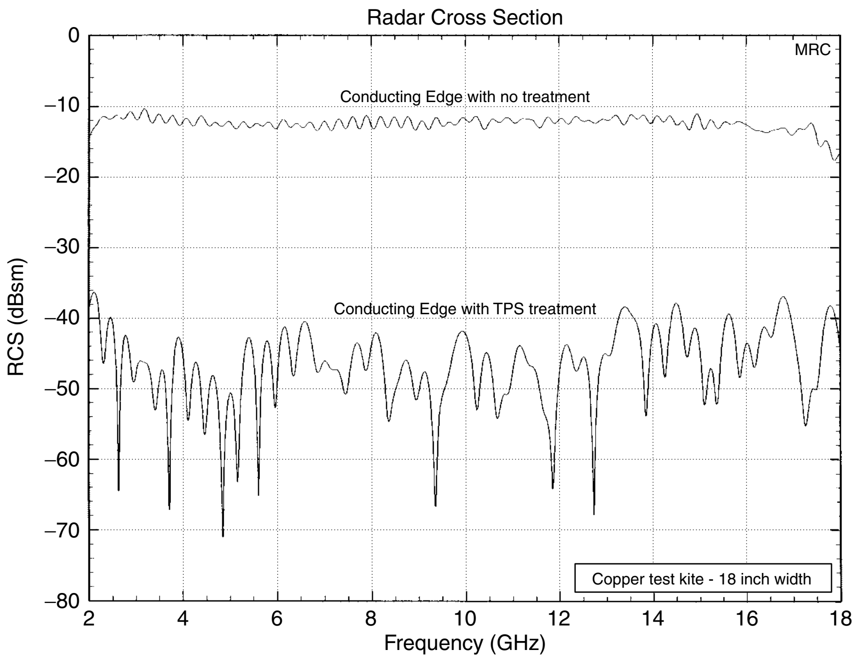


Fig. 9.21 RCS spectrum of an 18-inch wide metal edge terminated with a 12 inch long broad band Tapered Periodic Surface, 30° above grazing, E-parallel. (bare metal edge case also shown).

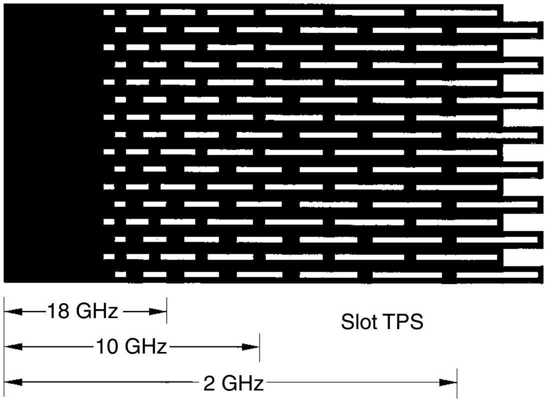


Fig. 9.22 The equivalent conductive length of a slot Tapered Periodic Surface decreases with increasing frequency.

The equivalent conductive length of a wire TPS increases with increasing frequency while that of a slot TPS decreases with increasing frequency. Figure 9.22 depicts this situation for a slot TPS. This "equivalent conductor" with its frequency dependent length can be used as a basic building block for many broad band antenna applications.

9.6.4 Application of TPS to a Pyramidal Horn Antenna

TPS may be used as extensions to the walls of a horn antenna. A conventional pyramidal horn antenna has relatively high side lobes and back lobes in the E-plane due to the strong edge diffraction at the aperture. TPS applied to the edges of the solid metal walls greatly reduces this diffracted energy. A slot TPS could also be designed such that the effective length (as well as the effective aperture size) of the horn decreases with increasing frequency thereby maintaining a frequency independent gain and beam width as well as low side lobes.

Figure 9.23 shows measured E-Plane patterns of two horn antennas. Both horns have the same flare angle and the same "effective length" producing approximately the same half power beam width. The pattern with higher side lobes is of a typical horn with metal walls. The other pattern in Fig. 9.23 is of a horn with walls terminated by TPS. Notice the vastly improved side and back lobe performance of the TPS horn.

9.6.5 Conclusions

It has been demonstrated that by gradually changing the size and/or spacing of the elements of an otherwise periodic surface, a smooth and continuous electromagnetic transition may be obtained. Not only does this transition exhibit very low EM field diffraction, but it also has the inherent ability to be frequency

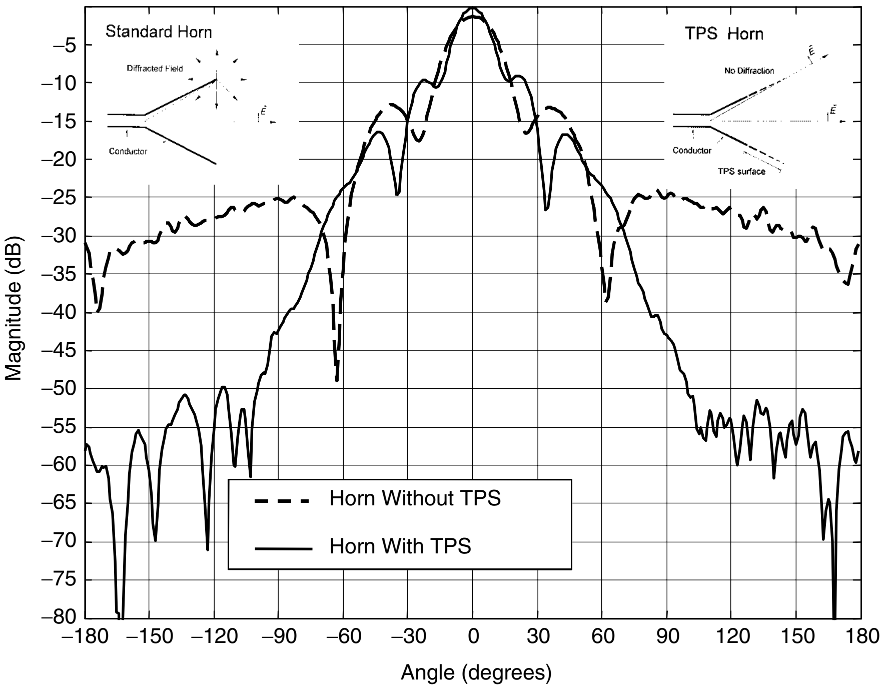


Fig. 9.23 E-plane pattern (at 6 GHz) of a horn antenna with and without Tapered Periodic Surface treatment of the E-plane walls.

independent. Other applications of TPS include reflector antenna edge treatment, aperture antennas with constant electrical dimensions (i.e. frequency independent performance), and low RCS test bodies with low diffracting edges.

9.7 CONCLUSIONS

Periodic structures are usually composed of identical elements positioned at equal spacings. In this chapter we have investigated surfaces made of two or more similar subarrays interlaced into each other in an arbitrary way. The subarrays were loaded with different load impedances, resulting in different resonance frequencies for the various subarrays.

For arrays of straight dipoles, this perturbation typically resulted in a slight upward shift of the original (unperturbed) resonant frequency. Furthermore, we obtained a lower, usually narrower, resonance frequency and a null located between the two resonant frequencies. No loss was observed at either resonant frequencies as long as no grating lobes were present in any of the subarrays. If that was not the case, we observed loss at both resonant frequencies and the null was more or less washed out.

These observations were typical for relatively small ratios between the two resonant frequencies (10–20%). For larger ratios like an octave or more, we demonstrated that arrays of large and small elements could be interlaced into each other and we could indeed obtain a dual-band structure, provided that;

1. Any harmonic resonance of the large elements should be moved away from the resonance region of the small elements.
2. Periodicity should be observed in the strictest of sense for both the small and the large elements.

In other words: Successful designs of dual- or multiband periodic surfaces is highly unlikely to occur by just leaving a computer to its own devices. A trained human brain is essential (as it is most of the time for even single-band structures).

We also demonstrated that insufficient modes on the elements could be troublesome. For example, if a four-legged loaded element did not incorporate the connector modes at the top and bottom (vertical polarization), we would observe anomalies for oblique angle of incidence. The two remaining modes would simply act as the reference elements for two subarrays with array spacings $d_x \neq D_x/2$, where D_x denotes the element spacings in the subarrays.

However, as illustrated in Section 9.6, kindly provided by Errol K. English, Mission Research Corporation, Dayton, Ohio, finite arrays where element length and spacing vary can be highly beneficial with many interesting applications.